YEAR 10 SCIENCE (GENERAL)
PHYSICAL SCIENCES
EXAMINATION REVISION - SEMESTER 2 2019

1. A top sprinter reaches 11.2 m/s within 3.15 s of the start of a race. Calculate the average acceleration.

$v = 11.2\ m/s$
$u = 0\ m/s$
$a = ?$
$t = 3.15\ s$

Take forwards as +ve.

$a = \frac{v-u}{t}$
$= \frac{11.2-0}{3.15}$
$= \underline{3.56\ m/s^2 \text{ forwards}}$

2. Calculate the final velocity of a car initially travelling at 10.0 m/s that accelerates at 1.4 m/s^2 for 8.0 s.

$v = ?$
$u = 10.0\ m/s$
$a = 1.4\ m/s^2$
$t = 8.0\ s$

Take forwards as +ve

$v = u + at$
$= 10.0 + (1.4)(8.0)$
$= \underline{21.2\ m/s \text{ forwards}}$

3. How long does it take for a F1 race car to accelerate at 8.70 m/s^2 from 150 km/h to 290 km/h as it enters a straight?

$v = 80.6\ m/s$
$u = 41.7\ m/s$
$a = 8.70\ m/s^2$
$t = ?$

Take forwards as +ve.

$a = \frac{v-u}{t}$
$\Rightarrow t = \frac{v-u}{a}$
$= \frac{80.6-41.7}{8.70}$
$= \underline{4.47\ s}$

4. A truck moving at 36 m/s is decelerated uniformly at 1.5 m/s^2. What is its final velocity after 5.0 s?

$v = ?$
$u = 36\,m/s$
$a = -1.5\,m/s^2$
$t = 5.0\,s$

Take forwards as +ve

$v = u + at$
$= 36 + (-1.5)(5.0)$
$= \underline{28.5\,m/s \text{ forwards}}$

5. A car of mass 1800 kg moving at 90 km/h applies its brakes, exerting a force of 2300 N for 3.5 s. What is the final velocity of the car?

$F = -2300\,N$
$m = 1800\,kg$
$a = ?$

Take forwards as +ve.

$F = ma$
$\Rightarrow a = \frac{F}{m}$
$= \frac{-2300}{1800}$
$= -1.28\,m/s^2$

$v = ?$
$u = 25.0\,m/s$
$a = -1.28\,m/s^2$
$t = 3.5\,s$

$v = u + at$
$= 25.0 + (-1.28)(3.5)$
$= \underline{20.5\,m/s \text{ forwards}}$

6. A car of mass 1530 kg is travelling at 70.0 km/h along Hepburn Avenue when the driver has to brake sharply to avoid a small child who suddenly appeared on a bike crossing the road.

(a) If it took 3.5 s to stop the car, calculate the average force exerted by the brakes.

$v = 0\,m/s$
$u = 19.4\,m/s$
$a = ?$
$t = 3.5\,s$

Take forwards as +ve.

$a = \frac{v-u}{t}$
$= \frac{0-19.4}{3.5}$
$= -5.54\,m/s^2$

$F =$
$m = 1530\,kg$
$a = -5.54\,m/s^2$

$F = ma$
$= (1530)(-5.54)$
$= \underline{-8476\,N \text{ backwards}}$

(b) The driver of the car was an engineer and had a hard-hat sitting on the back shelf. Describe what happened to the hard-hat in this situation and explain why it took place.

- Hard hat continues to move forward at 70.0 km/h initially.
- It has inertia due to the movement of the car.

7. (a) Explain the difference between ***mass*** and ***weight***. Give an example to help your explanation.

MASS: a measure of the amount of matter in an object.

WEIGHT: the force due to gravity acting on an object.

(b) "Space is great! You are weightless and it is easy to move stuff around." Comment on the accuracy of this statement.

- Not accurate.
- You "feel" weightless as there is no reaction force acting on you from the space ship you are in.
- Gravity acts everywhere in the universe, so every object has a weight force.

(c) The acceleration due to gravity on Neptune is 11.1 m/s^2. Calculate the weight of a 200 kg object on Jupiter.

$F_w = ?$

$m = 200 \text{ kg}$

$g = 11.1 \text{ m/s}^2$

$F_w = mg$

$= (200)(11.1)$

$= \underline{2220 \text{ N}}$

8. A 70.0 kg cyclist is applying a force of 150 N forwards but is working against a wind producing a friction force of 80 N opposing the movement.

(a) Determine the resultant force for this situation.

Take forwards as +ve.

$$\Sigma F = 150 - 80$$

$$= 70 \text{ N forwards}$$

(b) Calculate the acceleration of the cyclist.

$\Sigma F = 70$ N

$m = 70.0$ kg

$a = ?$

$$\Sigma F = ma$$

$$\Rightarrow a = \frac{\Sigma F}{m}$$

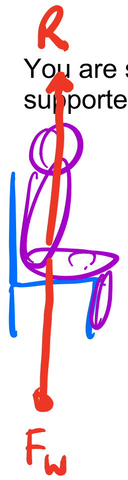
$$= \frac{70}{70.0}$$

$$= 1.0 \text{ m/s}^2 \text{ forwards}$$

(c) It is possible for the rider to move at constant velocity. In terms of the forces acting, explain how this is possible.

- Yes
- Apply a force of 80 N forwards so $\Sigma F = 0$.

9 . You are sitting on a seat at the moment. Explain why you don't fall through the chair and are supported by it.

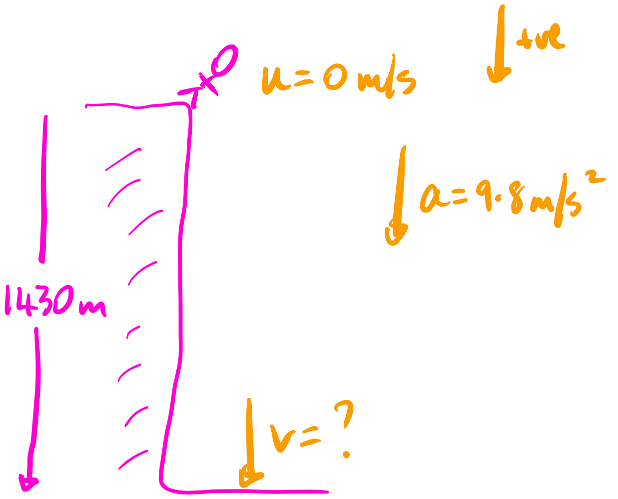


- The chair exerts a reaction force (R) upwards.
- R is equal and opposite F_w

$\therefore$ $\Sigma F = 0$ and the chair supports the person.

10. Steven enjoys BASE jumping off cliffs. He particularly enjoys a high cliff in Pakistan called Great Trango, which is 1340 m high.

(a) What is Steven's theoretical velocity after falling for 14.0 s, which is a fall of 1000 m?



$v = ?$
$u = 0\ ms^{-1}$
$a = 9.8\ m/s^2$
$t = 14.0\ s$

$v = u + at$
$= 0 + (9.8)(14.0)$
$= \underline{137.2\ m/s\ \text{down}}$

(b) Why is your answer to part (a) theoretical?

- This assumes no air resistance (friction).
- Friction exerts an opposing force so Steven does not accelerate at $9.80\ m/s^2$.

(c) Steven is actually moving at 45.0 m/s at this point when he releases his parachute. What force does it exert on Steven if he decelerates to 25.0 m/s in 1.4 s? Steven has a mass of 85.0 kg.

$v = 25.0\ m/s$ $\downarrow$ +ve
$u = 45\ m/s$
$a = ?$
$t = 1.4\ s$

$a = \dfrac{v - u}{t}$
$= \dfrac{25 - 45}{1.4}$
$= -14.3\ m/s^2$

$F = ?$
$m = 85.0\ kg$
$a = -14.3\ m/s^2$

$F = ma$
$= (85.0)(-14.3)$
$= \underline{-1216\ N\ \text{upwards}}$